

Temporal Difference Method

Similar to the MC method, temporal difference methods only require experiences, i.e., sample trajectories (sequence of states, actions, rewards).

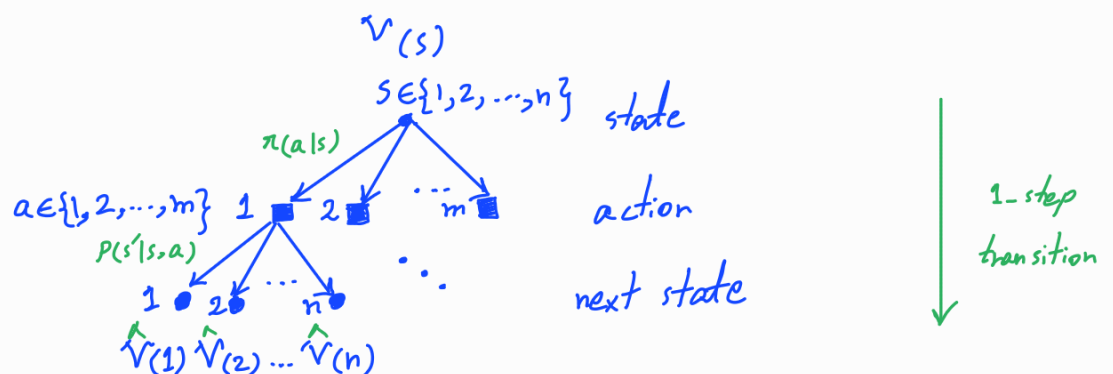
Model-free method

Temporal difference: Using dynamic programming over estimated values to learn online from incomplete episodes.

TD for policy evaluation: Given a stationary policy π , we want to compute

$$V^{\pi}(s) = \mathbb{E}_{\substack{A_t \sim \pi \\ S_{t+1} \sim P(\cdot | S_t, A_t)}} \left[\overbrace{\sum_{t=0}^T \gamma^t R(s_t, A_t)}^{\text{return}} \mid s_0 = s \right]$$

Idea: Estimate the expected return using the sample immediate reward and the current estimate of the future expected return value function).



Let $\hat{V}^\pi$ be the current estimate of the value function of π

$$\longrightarrow V^\pi(s) = \mathbb{E}_{\substack{A_t \sim \pi \\ s_{t+1} \sim P(\cdot | s_t, A_t)}} \left[\sum_{t=0}^T \gamma^t R(s_t, A_t) \mid s_0 = s \right]$$

$$= \mathbb{E}_{\substack{A_t \sim \pi \\ s_{t+1} \sim P(\cdot | s_t, A_t)}} \left[R(s_0, A_0) + \sum_{t=1}^T \gamma^t R(s_t, A_t) \mid s_0 = s \right]$$

Bellman consistency equation

$$V^\pi(s) = \mathbb{E}_{a, s'} [R(s, a) + \gamma V^\pi(s')]$$

$$= \mathbb{E}_{A_0 \sim \pi(s_0)} \left[R(s_0, A_0) + \underbrace{\gamma \mathbb{E}_{\substack{A_t \sim \pi \\ s_{t+1} \sim P(\cdot | s_t, A_t)}} \left[\sum_{t=0}^T \gamma^t R(s_t, A_t) \mid s_0 = s' \right]}_{V^\pi(s')} \right]$$

$$\longrightarrow \hat{V}^\pi(s) = \frac{1}{N} \sum_{i=1}^N (R(s, a^i) + \gamma \hat{V}^\pi(s'^i))$$

samples $\longleftarrow$ $\longleftarrow$ sample i

cannot be done online for $N > 1$
 $\downarrow$
 incremental computation of mean

Recall the MC update:

$$\hat{V}_{N(s)}^\pi(s) = \frac{1}{N(s)} \sum_{i=1}^{N(s)} G_i^s$$

$$\hat{V}_{N(s)+1}^\pi(s) = \hat{V}_{N(s)}^\pi(s) + \frac{1}{N(s)+1} (G_{N(s)+1}^s - \hat{V}_{N(s)}^\pi(s))$$

$$\hat{V}^\pi(s) \leftarrow \hat{V}^\pi(s) + \alpha (G - \hat{V}^\pi(s))$$

$\longleftarrow$ for all states visited in the episode
 $\longleftarrow$ learning rate / step size
 $\longleftarrow$ MC target
 $\longleftarrow$ MC error

$\swarrow$ equivalent for certain choice of $\alpha = \frac{1}{N(s)+1}$

TD(0) update:

$$\hat{V}^\pi(s) \leftarrow \hat{V}^\pi(s) + \alpha (R(s, a) + \gamma \hat{V}^\pi(s') - \hat{V}^\pi(s))$$

$\longleftarrow$ for the state just visited
 $\longleftarrow$ learning rate / step size
 $\longleftarrow$ TD target
 $\longleftarrow$ TD error

TD (0) method

Iterative method:

- Initialize $\hat{V}^\pi$

- Repeat

- Sample $s_0 \in S$

- Repeat until the episode terminates

- Take action a according to $\pi(a|s)$ at s

- Observe reward $R(s, a)$ and next state s'

- $\hat{V}^\pi(s) \leftarrow \hat{V}^\pi(s) + \alpha [R(s, a) + \gamma \hat{V}^\pi(s') - \hat{V}^\pi(s)]$

- Update the current state $s \leftarrow s'$

$$\begin{bmatrix} \hat{V}^\pi_{(1)} \\ \hat{V}^\pi_{(2)} \\ \vdots \\ \hat{V}^\pi_{(n)} \end{bmatrix}_{|S| \times 1} \xrightarrow{s=2 \xrightarrow{\pi} a=1 \xrightarrow{P} s'=4, r=R(2,1)} \hat{V}^\pi_{(2)} \leftarrow \hat{V}^\pi_{(2)} + \alpha (r + \gamma \hat{V}^\pi_{(4)} - \hat{V}^\pi_{(2)})$$

TD for policy optimization: We want to find an (approximately) optimal policy π^* through iterative policy evaluation and policy improvement.



Idea:

- Evaluate the policy by estimating the Q function using the TD method in an online fashion.
- Improve the policy using the estimated Q function in a greedy manner.

SARSA $\rightarrow s, a, r, s', a'$

Iterative method:

- Initialize $\hat{Q}$ s.t. $\hat{Q}(s, a) = 0$ for terminal states

- Initialize π according to $\hat{Q}$ (e.g., ϵ -greedy)

- Repeat

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|A|} & \text{if } a = \underset{a \in A}{\operatorname{argmax}} \hat{Q}(s, a) \\ \frac{\epsilon}{|A|} & \text{otherwise} \end{cases}$$

- Sample $s_0 \in S$

- Select action a according to $\pi(a|s)$ at s_0

- Repeat until the episode terminates

- Take action a

- Observe reward $R(s, a)$ and next state s'

- Select action a' according to $\pi(a'|s')$ at s'

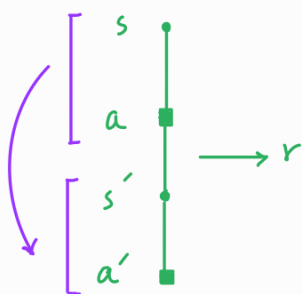
policy evaluation • $\hat{Q}(s, a) \leftarrow \hat{Q}(s, a) + \alpha [\underbrace{R(s, a) + \gamma \hat{Q}(s', a')}_{\text{TD target}} - \hat{Q}(s, a)]$

policy improvement • Build $\pi(a|s)$ according to $\hat{Q}$ (e.g., ϵ -greedy)

- Update the current state and the current action

$$s \leftarrow s'$$

$$a \leftarrow a'$$

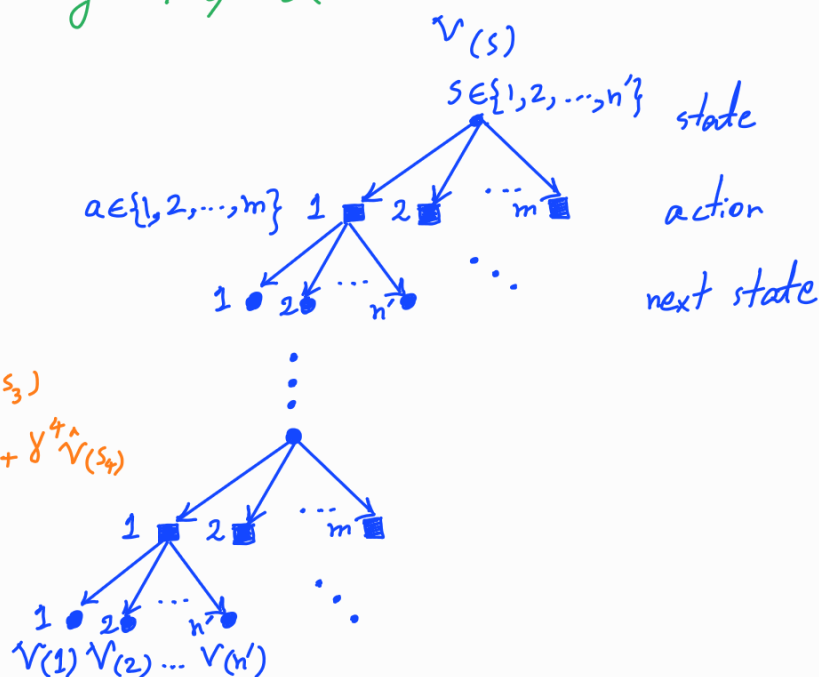


Note [n-step TD method]: Estimate the expected return using n sample immediate rewards and the current estimate of the future expected return (value function) after these n steps.

$$\hat{V}^{\pi}(s_t) \leftarrow \hat{V}^{\pi}(s_t) + \alpha (G_t^{t+h} - \hat{V}^{\pi}(s_t))$$

learning rate / step size
TD target
TD error
n-step return

$$G_t^{t+h} = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots + \gamma^{n-1} r_{t+n-1} + \gamma^n \hat{V}^{\pi}(s_{t+h})$$



n-step transition

$$\hat{V}(s_0)$$

$$G_0^1 = r_0 + \gamma \hat{V}(s_1)$$

$$G_0^2 = r_0 + \gamma r_1 + \gamma^2 \hat{V}(s_2)$$

$$G_0^3 = r_0 + \gamma r_1 + \gamma^2 r_2 + \gamma^3 \hat{V}(s_3)$$

$$G_0^4 = r_0 + \gamma r_1 + \gamma^2 r_2 + \gamma^3 r_3 + \gamma^4 \hat{V}(s_4)$$

$$\bar{G} = \frac{1}{4} \sum_{i=1}^4 G_0^i$$

Note [TD(λ) method]: Updates the expected return using a weighted average of n -step returns over varying n .

$$\hat{V}^{\pi}(s_t) \leftarrow \hat{V}^{\pi}(s_t) + \alpha (G_t^{\lambda} - \hat{V}^{\pi}(s_t))$$

learning rate / step size
TD target
TD error
weighted average of n-step returns

$$G_t^{\lambda} = (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_t^{t+n}$$

$\lambda \in [0, 1]$

if $\lambda = 0 \rightarrow \text{TD}(0)$
 if $\lambda = 1 \rightarrow \text{MC}$

$$G_t^{\lambda} = (1-\lambda) \sum_{n=1}^{T-t-1} \lambda^{n-1} G_t^{t+n} + \lambda \sum_{n=T-t}^{T-1} G_t^n$$

* Backward view of TD(λ) using the idea of eligibility traces provides an incremental approach to approximate the forward view.

Backward view of $TD(\lambda)$: An incremental approach to approximate the forward view.

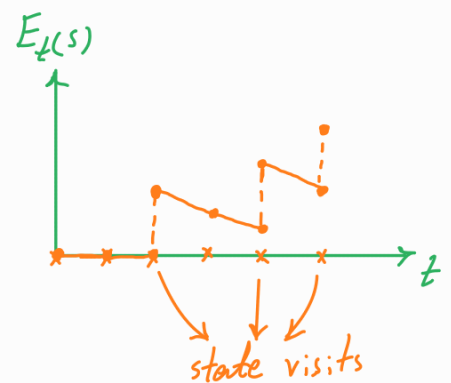
Eligibility trace

$E_t : S \rightarrow \mathbb{R}$ is the eligibility trace of state s at time t

accumulating eligibility trace

$$\begin{cases} E_0(s) = 0 \\ E_t(s) = \gamma \lambda E_{t-1}(s) + \mathbb{1}(S_t = s) \end{cases}$$

γ trace-decay parameter
 λ time progress decreases eligibility
 $\mathbb{1}(S_t = s)$ state visit increases eligibility



Idea: Update the estimated values of states according to how eligible the states are for being updated.

$TD(\lambda)$ backward update:

$$\delta_t = R_t(S_t, A_t) + \gamma \hat{V}^{\pi}(S_{t+1}) - \hat{V}^{\pi}(S_t)$$

δ_t is the TD error

$$\hat{V}^{\pi}(s) \leftarrow \hat{V}^{\pi}(s) + \alpha \delta_t E_t(s)$$

for all $s \in S$

Online TD(λ) for policy evaluation

Iterative method:

- Initialize $\hat{V}^\pi$ s.t. $\hat{V}^\pi(s) = 0$ for terminal states
- Repeat
 - Initialize $E(s) = 0$ for all $s \in \mathcal{S}$
 - Sample $s_0 \in \mathcal{S}$ at $t = 0$
 - Repeat until the episode terminates
 - Take action A_t according to $\pi(a|s)$ at s_t
 - Observe reward $R(s_t, A_t)$ and next state s_{t+1}
 - $\delta \leftarrow R(s_t, A_t) + \gamma \hat{V}^\pi(s_{t+1}) - \hat{V}^\pi(s_t)$
 - For $s \in \mathcal{S}$:
 - $E(s) \leftarrow \gamma \lambda E(s) + \mathbb{1}(s_t = s)$
 - $\hat{V}^\pi(s) \leftarrow \hat{V}^\pi(s) + \alpha \delta E(s)$
 - $t \leftarrow t + 1$

SARSA (λ)

Iterative method:

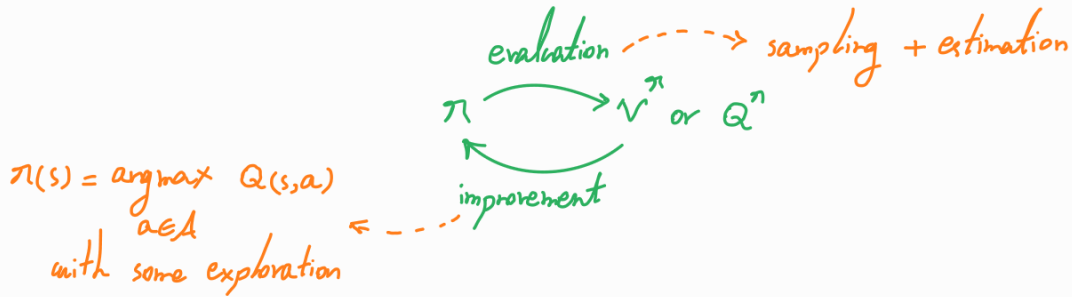
- Initialize $\hat{Q}$
- Initialize $\pi(a|s)$ according to $\hat{Q}$ (e.g., ϵ -greedy)
- Repeat
 - Initialize $E(s, a) = 0$ for all $s \in S, a \in A$
 - Sample $s_0 \in S$ at $t = 0$
 - Select action A_0 according to $\pi(a|s)$ at s_0
 - Repeat until the episode terminates
 - Take action A_t
 - Observe reward $R(s_t, A_t)$ and next state s_{t+1}
 - Select action A_{t+1} according to $\pi(a|s)$ at s_{t+1}
 - $\delta \leftarrow R(s_t, A_t) + \gamma \hat{Q}(s_{t+1}, A_{t+1}) - \hat{Q}(s_t, A_t)$
 - For $s \in S, a \in A$:
 - $E(s, a) \leftarrow \gamma \lambda E(s, a) + \mathbb{1}(s_t = s, A_t = a)$
 - $\hat{Q}(s, a) \leftarrow \hat{Q}(s, a) + \alpha \delta E(s, a)$
 - Build $\pi(a|s)$ according to $\hat{Q}$ (e.g., ϵ -greedy)
 - $t \leftarrow t + 1$

policy evaluation

policy improvement

MC vs. TD

Both algorithms are based on generalized policy iteration.



MC	$\hat{V}^\pi(s_0) \leftarrow \hat{V}^\pi(s_0) + \alpha \left(\overbrace{r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^T r_T}^{G = G_0^T} - \hat{V}^\pi(s_0) \right)$
1-step TD(0)	$\hat{V}^\pi(s_0) \leftarrow \hat{V}^\pi(s_0) + \alpha \left(\overbrace{r_0 + \gamma \hat{V}^\pi(s_1)}^{G_0^1} - \hat{V}^\pi(s_0) \right)$
2-step TD(0)	$\hat{V}^\pi(s_0) \leftarrow \hat{V}^\pi(s_0) + \alpha \left(\overbrace{r_0 + \gamma r_1 + \gamma^2 \hat{V}^\pi(s_2)}^{G_0^2} - \hat{V}^\pi(s_0) \right)$
⋮	⋮
n-step TD(0)	$\hat{V}^\pi(s_0) \leftarrow \hat{V}^\pi(s_0) + \alpha \left(\overbrace{r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^{n-1} r_{n-1} + \gamma^n \hat{V}^\pi(s_n)}^{G_0^n} - \hat{V}^\pi(s_0) \right)$
TD(λ)	$\hat{V}^\pi(s_0) \leftarrow \hat{V}^\pi(s_0) + \alpha \left(G_0^\lambda - \hat{V}^\pi(s_0) \right)$

Bias-Variance Trade-off in Estimation:

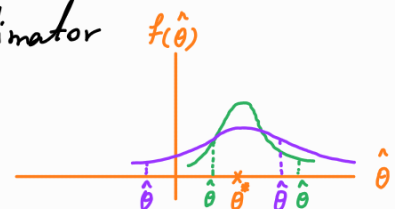
θ^*

$\xrightarrow{D} x_i$

$\hat{\theta} = \text{est}(\{x_i\}_{i=1}^N) \approx \theta^*$

Example: $\theta^* = E_D[X] \quad \hat{\theta} = \frac{1}{N} \sum_{i=1}^N x_i$

$$\begin{cases} \text{Bias} = E_D[\hat{\theta}] - \theta^* & \text{if } E_D[\hat{\theta}] = \theta^* \rightarrow \text{unbiased estimator} \\ \text{Variance} = E_D[(\hat{\theta} - E_D[\hat{\theta}])^2] \end{cases}$$



A good estimator has low bias and low variance.

Example: In certain supervised learning settings, it holds that

$$\text{Generalization error} = \text{bias}^2 + \text{variance} + \text{data noise}$$

However, there are often some trade-offs between reducing bias and variance.

* MC and TD methods both estimate the value functions $\hat{V}^\pi$ and $\hat{Q}^\pi$.

MC-based estimator is unbiased but potentially with high variance.

TD(0)-based estimator is biased but potentially with low variance.

TD(λ)-based estimator can balance bias and variance through parameter λ .